

Introduction

CS 350

Dr. Joseph Eremondi

Last updated: December 18, 2024

Course Overview

Course Objectives

To learn:

- Functional programming

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 - Recursion

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 - Evaluation

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 - Evaluation

To change how you *think* about programming

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 - We'll follow loosely

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 - Announcements

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 - Assignments and Handin

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 - EXCEPTION: when you can't ask your question without revealing your solution to the assignment

- Location: RIC 317

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 - Take the elevator, go across the bridge and through the door

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Grading Scheme

- 35% assignments

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- 15% midterm

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- 10% Participation

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- Assignment 0 is 1% (counts as 2 lectures)

Assignment 0

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CS350 Handin



Assignments 1-6

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 - Last two are “optional” if you’re happy with your marks

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- **READ THE TEST RESULTS** to know what you need to fix

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- It's better to submit code that fails some tests than to submit code with type or syntax errors

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 - Copying code will probably get you a 0

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Don't set yourself up for failure on the exams

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Programming language genealogy and design.

Imperative, functional, and object-oriented language paradigms.

Context-free grammars and syntax trees.

Data types, control structures, exception handling, data abstraction, information hiding, and non-determinism.

Program representation, translation, and execution.

Functional programming: advantages, constructs, closures, and higher-order operations.

Parallel programming.

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- *First half of the class: Programming in Flit*

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- Am I going to use Racket in Industry?

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 - No

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 - Then you'll be able to learn whatever language you need in industry

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 - Yes. **This is a programming languages class.**

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- Do you really expect me to learn a new language in just one term?
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 - The language we're learning is small (really just functions and pattern matching)

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 - But I can teach you what they're made of

- How to think about programs in a rigorous way

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 - Equations with programs

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 - How the shape of a problem can guide to its solution
 - Using types to communicate properties of code
- This is radically different from the kind of coding you have done so far